

RockSat-X 2024 Final Report

DAEDALUS II



Ethan Buchanan, Jackson Berreth, Morgan Erler, Jonathan Herberger, Juliette Haggith,
Riley Mark

Dr. Dan Lawrence, Dr. Autumn Pratt

Northwest Nazarene University

30 September 2022

1.0 Mission Statement

Current spacefaring robotic systems struggle to adapt to varying tasks and environments. Intelligent robotic systems have the potential to assist astronauts and solve challenging problems such as space debris removal, collision avoidance, and in-orbit spacecraft servicing and assembly. Current robotic systems used in space require manual control, which will prove difficult for distant uncrewed missions where communication times make remote control impractical. Remote control will also hinder the creation of large robotic workforces, as the necessity for human involvement would be too great. This technology could be revolutionary in the aerospace industry as an intelligent, dexterous system could learn to perform tasks in space and would decrease the need for human intervention. This work aims to develop adaptive robotic arms that train in simulation and then complete a complex task in the real-world space environment. For this project, the complex task involves one robotic arm that catches a ball in space.

2.0 Mission Requirements and Description

The NNU Rocksat-X team plans to iterate on our prior work completed on the Daedalus One mission. The team plans to develop a robotic arm capable of performing complex tasks in microgravity. To test the robotic arm, a series of objects will be launched from a deployment system. The robotic arm will attempt to catch the ejected objects using a camera attached to the end effector. This robotic arm will attempt to catch objects launched from a deployment subsystem. With the culmination of prior years' research and lessons learned, the team plans to update our experimental setup.

Due to increased demand for experiment slots, the team decided to downsize the experiment to a half-cam. With this choice, many design considerations needed to be changed. The team decided to replace the two-arm system with a singular arm and a ball launcher. With these changes, the team hopes all subsystems will comfortably fit within the usable volume. The object deployment method will strive for a safe and highly predictable trajectory.

Other requirements include determining an inverse kinematic particular solution for the custom robotic arm; one that can accurately and repeatedly relate the robotic arm's encoder positions to the x y z vector location of the arm. A new vision system that can accurately determine an object's location while using few computational resources.

The payload must be completely enclosed in the usable payload volume. The system did not use more than 28 V, and had a max electrical charge of 1 Ah. These requirements were considered when choosing the different electrical components, creating the software, and designing the mechanical system. The self-imposed requirements included having an arm deployed, a ball launched and caught, and a video recording of the entire process being saved.

3.0 Payload Design

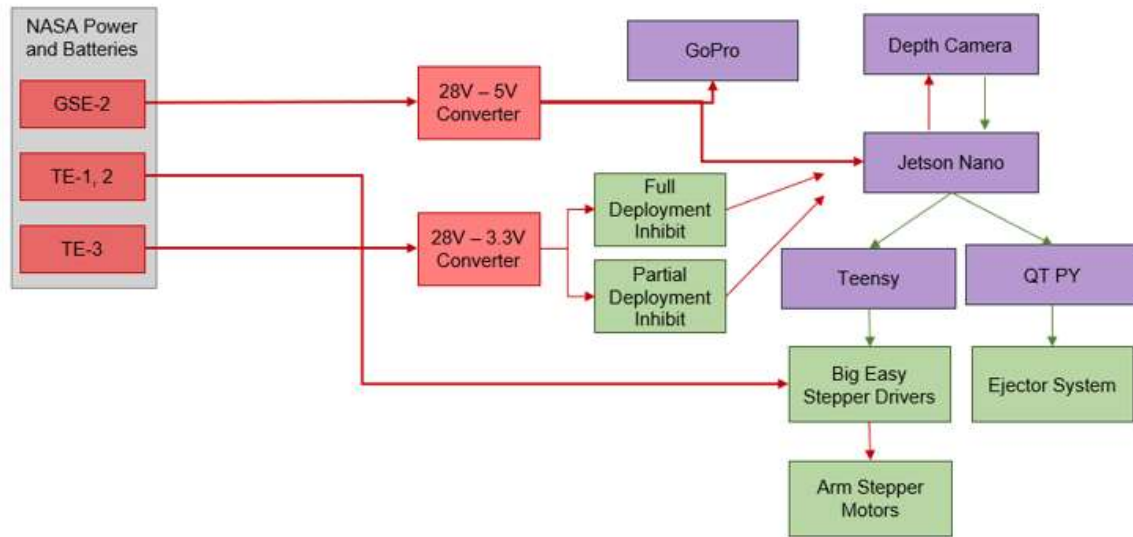


Figure 1 - Functional Block Diagram

DAEDALUS II Design Overview

This payload was designed to incorporate one robotic arm, one ball launcher, three balls, and all the necessary electrical components for the DAEDALUS II project. Figure 2 shows a rendering of the final design of the payload as it would look while operating in space.

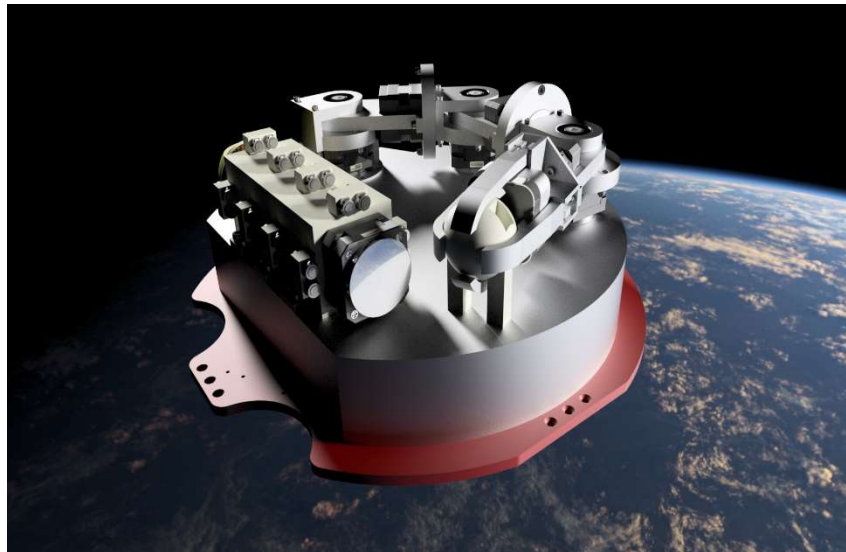


Figure 2: DAEDALUS II Deployment CAD Rendering

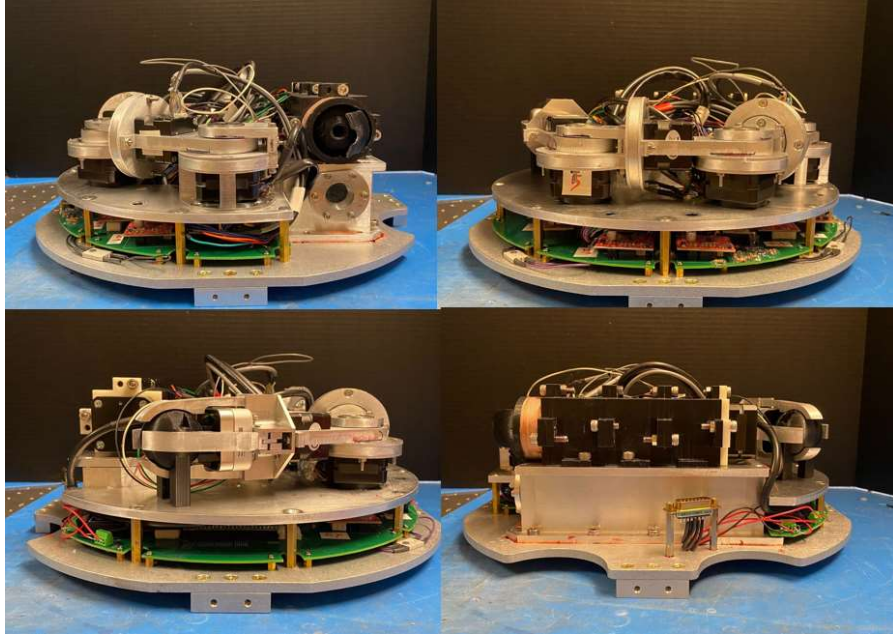


Figure 3: DAEDALUS Pre-Flight Orthogonal Views



Figure 4: DAEDALUS Pre-Flight Isometric Views

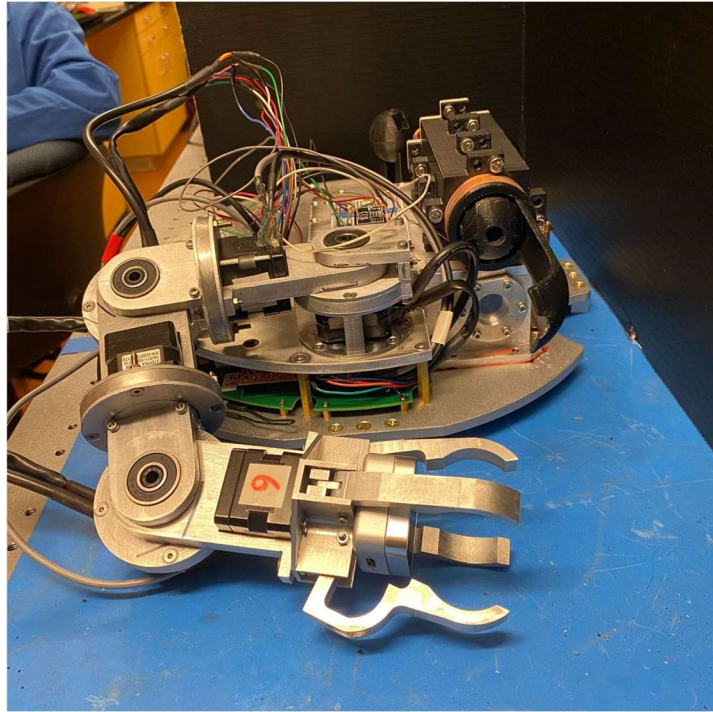


Figure 5: DAEDALUS Pre-Flight Deployment View

Robotic Arm Design

From the Daedalus I experiment, many changes were made to the arm design. Due to the usable space constraints, the design was adjusted from a two-arm system to one arm to fit within the half-cam volume. The arm needed to be designed to fit within this compact space, unfold efficiently, and perform a series of maneuvers to catch and release a ball. The arm design was adjusted from a six-axis arm to a five-axis arm. This allowed the arm to be mounted horizontally on a baseplate and then unfold from the side. The final arm was made of aluminum and included multiple stepper motors to control the arm's movements. Each joint continued to use planetary gearboxes to compensate for the amount of torque the arm would experience.

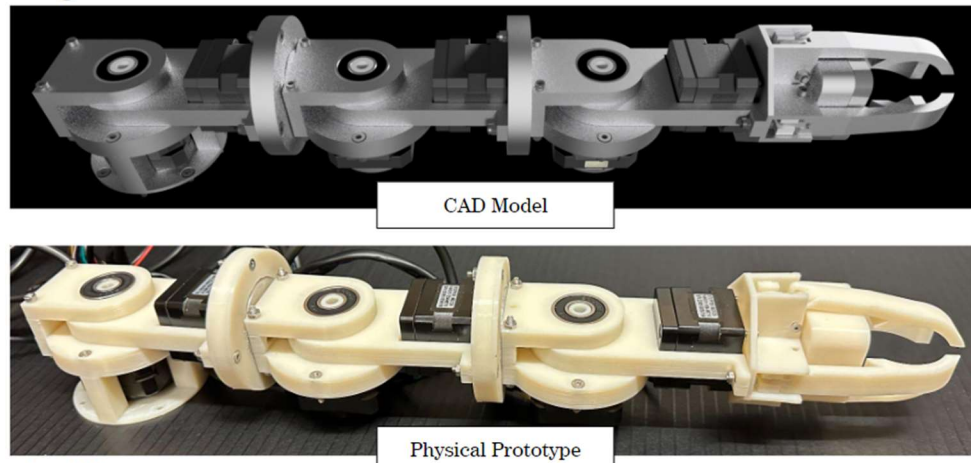


Figure 6: DAEDALUS II Concept and Physical Prototype

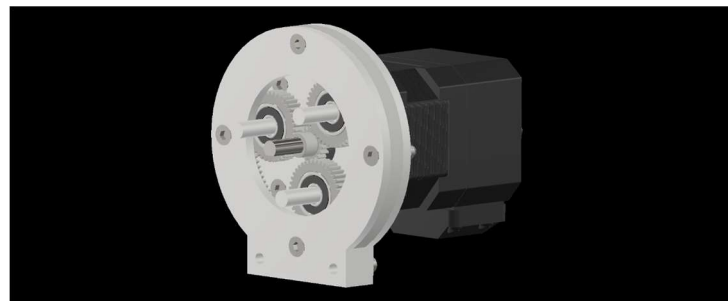


Figure 7: Planetary Gearbox CAD Model

The robotic arm featured an independently-actuated robotic gripper that used a single stepper motor and threaded rod to animate four gripper links. The gripper was designed to interact with a ball of known dimensions. When unpowered, the ACNE thread functioned as a locking mechanism, securing itself to the payload deck for launch and re-entry.

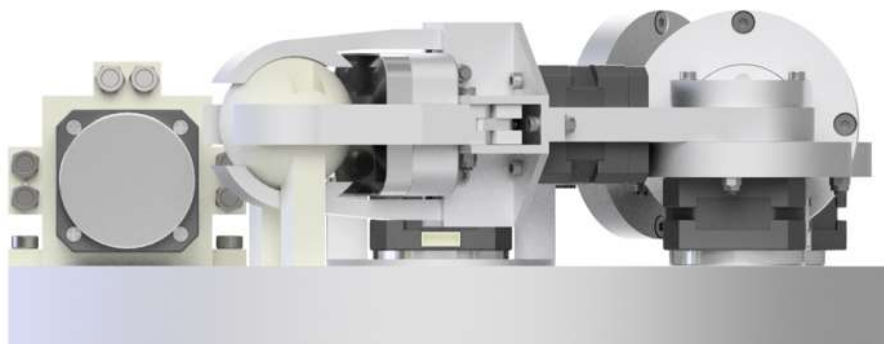


Figure 8: Gripper/Gripper Mount CAD Model

EMBER Design

To precisely and repeatedly launch deployables the EMBER system was developed. This system used an electromagnet as the mechanism of launch to avoid stored energy. Using a Stepper motor and an ACNE thread, the balls were moved into position within the electromagnet and then deployed after a set time. A guide rail was implemented to position the balls outside the usable payload volume, where the arm could safely attempt a catch.

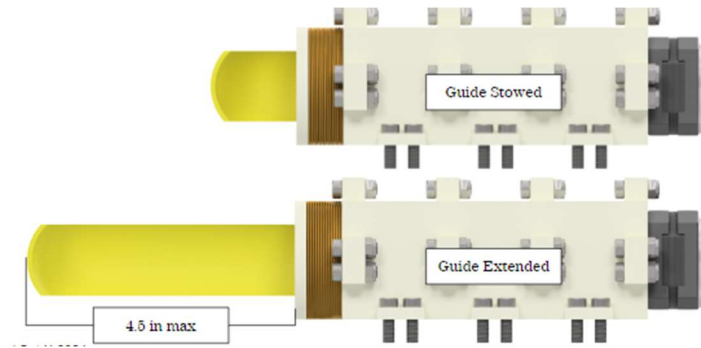


Figure 9: EMBER prototype and CAD Model

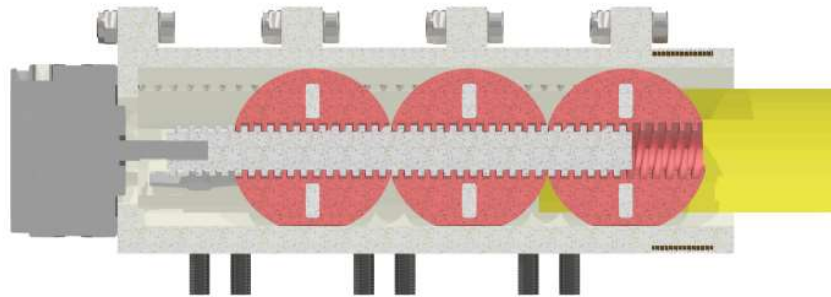


Figure 10: EMBER sectional view

The system follows the block diagram shown. Electrical power is shown with red arrows, analog data flows with the green arrow, and the blue arrows indicate physical movement through the system. The EMBER system is controlled by the QT Py, which receives a signal from the Daedalus computer to eject, and then it runs through a two-step ejection process. First, the motor will move for a set amount of time, pushing the ball into the ejection zone. Then the current will be fed through the solenoid ejecting the ball out of the system.

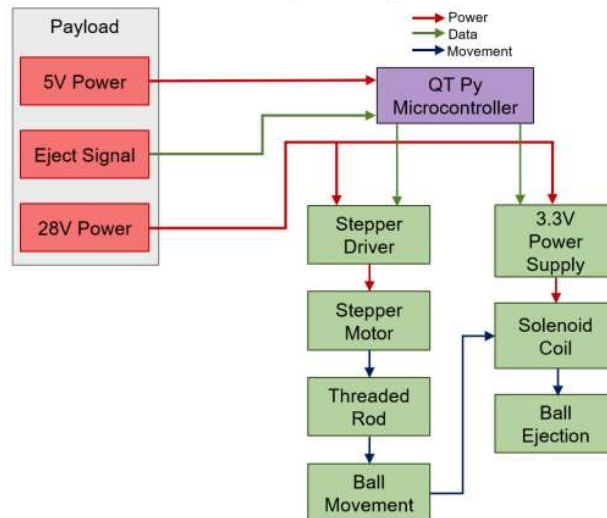


Figure 11: EMBER System Diagram

Electrical System

Iterating on previous years' design, a new printed circuit board (PCB) was developed. Using Fusion360, a single board was designed to house all microcontrollers, effectively facilitating communication between subsystems. The board was attached directly to the baseplate using small standoffs. This ensures that there is minimal movement during vibration testing and flight. The final iteration of the PCB effectively wraps around the dry box, the circuit diagram was mapped to the remaining usable experiment area.



Figure 12 - First Electrical System

Features of the board include test pads, where the voltage of power traces can be measured, and capacitor mounts, where a capacitor can be soldered to reduce noise and interference in the power lines. After rigorous testing, it was determined that no capacitors were needed to reduce noise. Header pins were used to connect both the BigEz stepper motor controllers and Teensy to the board. This feature allowed for quick replacement of microcontrollers in the event of a malfunction. Similarly, 9-pin D-Sub connectors were used to connect individual stepper motors and respective encoders with the associated microcontroller, see Figure 13. In the event of a motor burn out, components can be replaced without resoldering the board.

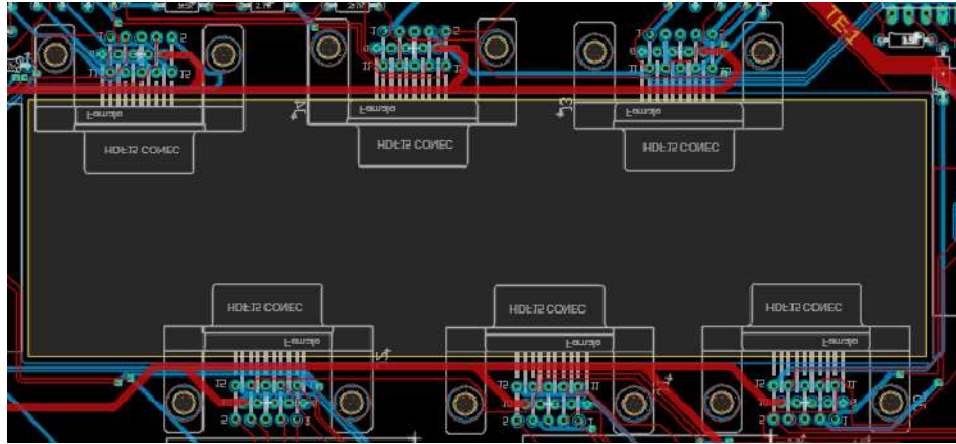


Figure 13 – D-Sub Stepper Motor Connectors

Power distribution enters the board at the bottom left of Figure 12. GSE power enters the board and is immediately stepped down to 5 volts using a 28V – 5-V converter. This line then sends power to the Jetson Teensy and GoPro. Power from TE1 and TE2 is immediately sent to all 6 BigEz Stepper Motor controllers.

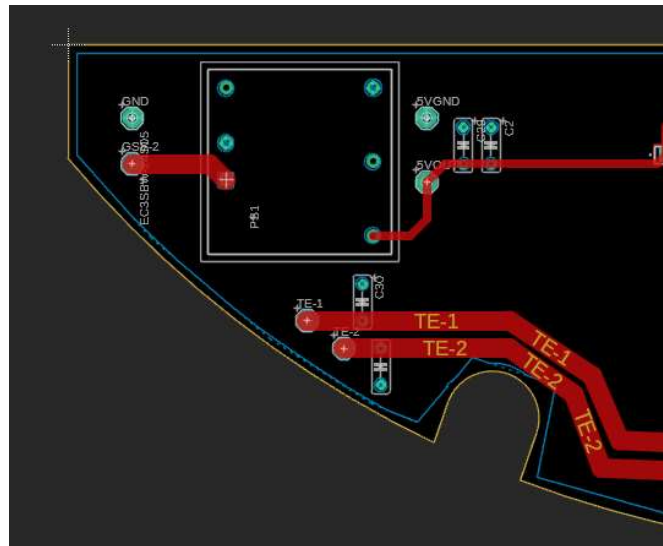


Figure 14 – Electrical Power Distribution

At the bottom of the board, the Teensy is mounted. When a movement command is sent from the Jetson, the Teensy determines the desired speed using a PI controller. A signal is sent to the BigEz microcontrollers and then to stepper motors through the D-Sub connectors. As the motor moves Using an op-amp circuit, the encoder values from the four redundant lines are compared, and then the value is sent to the Teensy.

A second board was developed to control the inhibitory circuitry. Due to late development, this circuit was built using a solderable breadboard. The circuit diagram in Figure 15 outlines the inhibit circuitry. Winchester pins were used to create breaks in the circuit. When a low was read on the GPIO board of the Jetson, the associated inhibit procedure was performed.

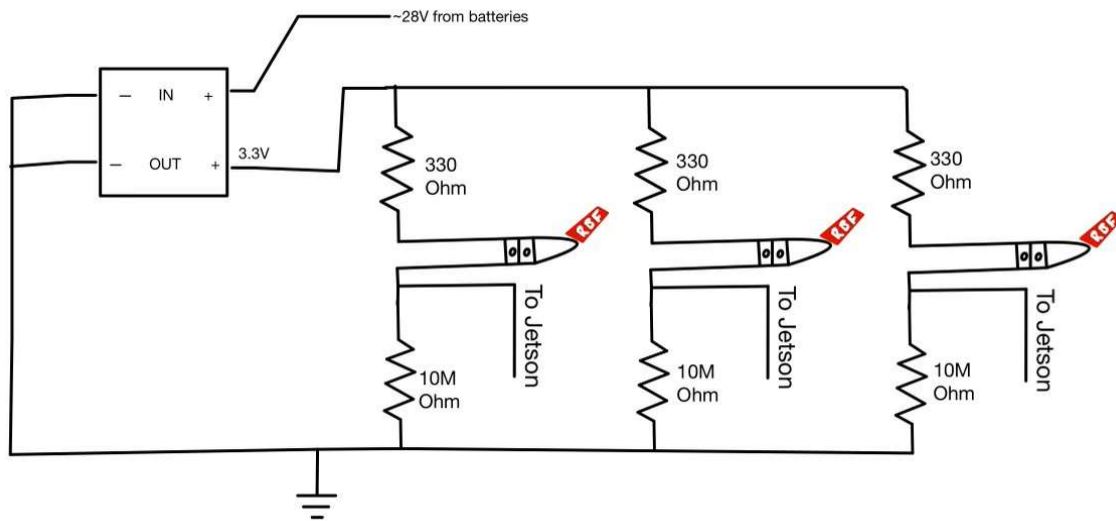


Figure 15 – Inhibit Board Circuit Diagram

Payload Command System Software

The payload requires an overhead system to control each of the subsystems. Named the payload command system (PCS), this system is executed on the main computer, an Nvidia Jetson. The PCS is a ROS service in which each payload subsystem queries to indicate its current state and to receive commands such as activation, deactivation, and shutdown. Outlined in Figure 16 the blue lines indicate a success or presence of multiple choices while orange lines indicate a failure.

When GSE is activate, the Jetson Nanos receive power and begin to boot up. During the bootup process, the PCS begins as a system service. At the beginning of the PCS, a check is run to determine the status of the TE lines. The system waits until this signal has been received, switching between the TE detection state and the wait for the TE delay state until the signal is received.

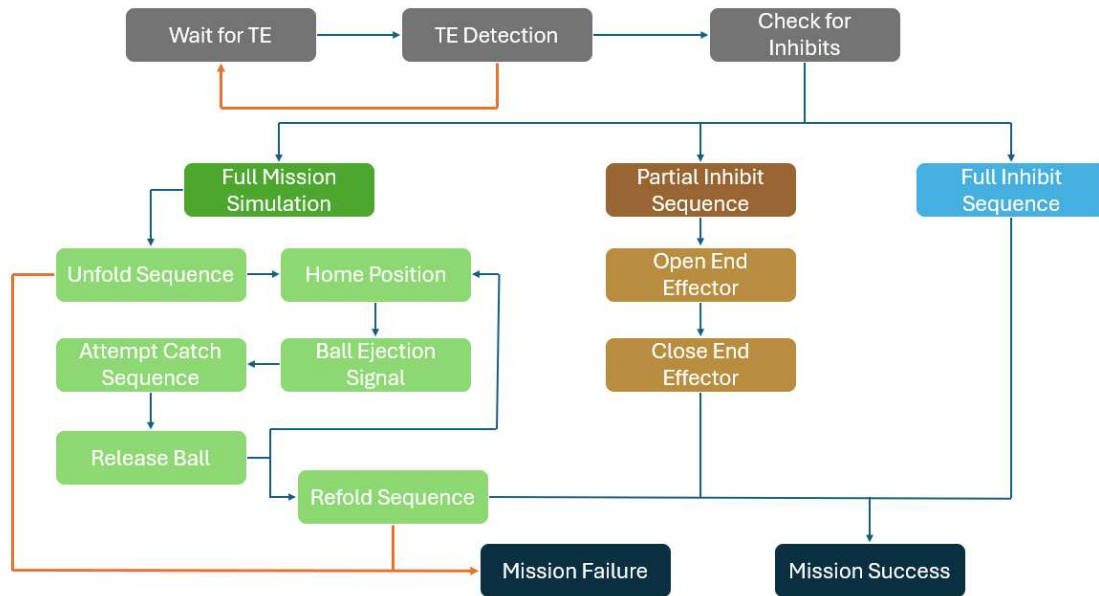


Figure 16 - Master Payload State Machine

At T+110, the first TE signal is sent, informing the experiment that the rocket has spun down. Once the first TE event is registered, the system checks for inhibits. The PCS system detects the inhibit board procedure using the Jetson's GPIO board. Depending on the combination of high and low signals, one of the three procedures is enacted. If the payload is fully inhibited, the system will exit, and all processes will cease. At this time, the stepper motors maintain their position and wait for power shutoff. If the partial procedure is called, the payload latches and unlatches to its fixed position on the payload. The combinations of these two tests allow the team to test the functionality of all arm joints without running a full mission sequence.

If the full mission sequence is called, the arm unlatches from its mount and begins unfolding; if it fails to reach the home position code, the system exits. At this point the arm will hold its current position until power down. In previous years, the team created a sequence to attempt refolding.

Once the home position is reached, the arm will begin the catch sequence. A signal is sent to the EMBER system to launch a ball. After a delay, the arm will attempt to catch the ball. Due to an issue with the vision system, this was done through a predetermined sequence of locations instead of using inverse kinematics. This is explained further in the computer vision section of payload design. Once a catch is attempted, the arm moves to release the object on a known trajectory. This process is repeated for each deployable, and then the arm refold sequence is called. This sequence steps through the unfolding sequence's positional code backward. The arm should end by latching to the fixed position.

Camera System

The payload features one camera solely for viewing catch attempts. This camera was mounted within the dry box facing the expected location of the ball launch and ball catch attempt. Due to the team's history with SD card file corruption, the GoPro camera platform was used due to its reliability and file repair system. Using the GoPro labs firmware QrCode generator, the camera was programmed to save footage in 15-second intervals, thereby avoiding loss of footage on power shutdown. While the camera can run on lithium-ion batteries, the team decided to power the camera through the main board to avoid unnecessary stored energy.

Computer Vision Software

To determine the position of the ball a color segmentation filter was used. Images of the deployable were taken, and then, using the MATLAB color threshold toolbox, the HSV values of the ball were determined. Using these values, a mask could be produced to identify the ball's location. In practice, the Intel RealSense 405 camera would snap an image and feed it to the main computer. Segmentation would be performed, and the centroid of the ball would be mapped. Using the depth camera's determined point cloud, the vector location of the centroid would be found relating the end of the arm's position to the ball.

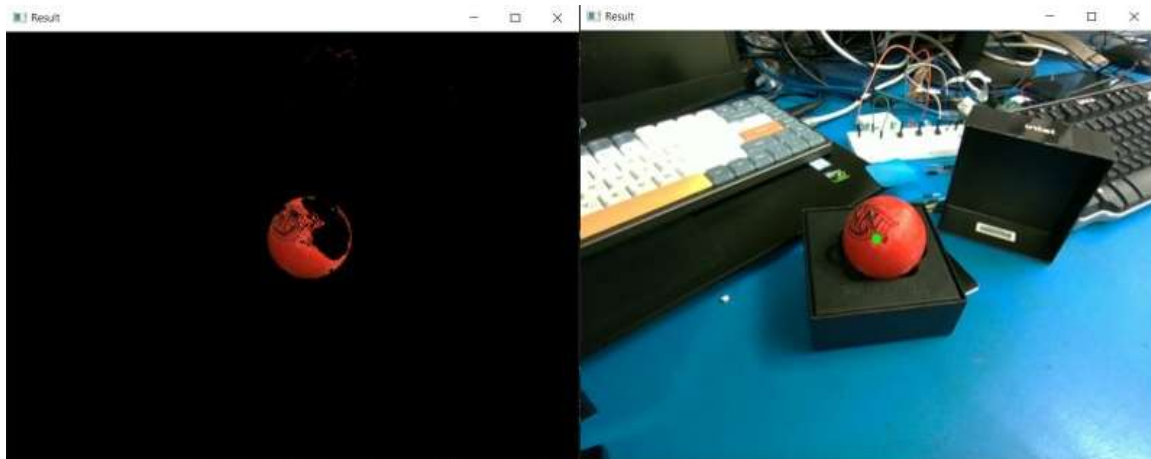


Figure 17 – Color Mask Segmentation and Determined Centroid

The arm would then move to this position using predetermined inverse kinematic equations. This process would cycle in a loop until the ball blocked the camera's vision. Due to time constraints, the inverse kinematics of the arm was never determined, leaving this system unusable. Instead, a choreographed set of positions for each joint was determined with the expected path of a ball.

4.0 Student Involvement

Project DAEDALUS was a student lead mission that consisted of seniors working to fulfill their design project requirements, paid research students, and volunteers. Members of the team were assigned different tasks to complete based on their experience and expertise.

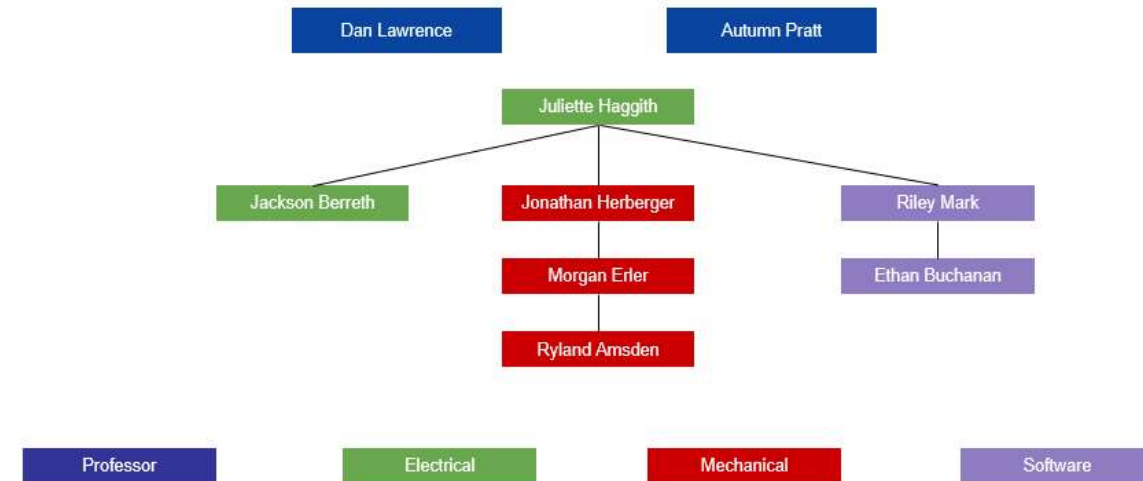


Figure 18 - MARSHA Organizational Chart

5.0 Testing Results

The DAEDALES system underwent significant testing throughout the design process to ensure that it was capable of surviving the launch process and was able to perform the necessary maneuvers to successfully carry out the experiment. Each subsystem was tested on its own before being integrated with the rest of the experiment.

EMBER Subsystem Testing

Repeated testing of the deployment method was tested to ensure consistent ejection. In testing a reloading procedure was created ensuring the greatest opportunity for success. In testing it was determined that the electrical system functioned nominally. The guide rail released consistently the motor positioned the deployables correctly and the solenoid launched the objects.

To ensure the internal cam would not shift and preemptively release, the subsystem was repeatedly shaken violently on the z axis. When powered, the stepper motor attached to the cam performed exceptionally at resisting movement in the z axis.

Robotic Arm Subsystem

Current draw was tuned for each joint of the arm. Using the potentiometers present on the joint microcontrollers each joint was allotted current. Total current draw was kept under 3 Amps at 28 volts with 1.5 Amps being split between TE1 and TE2. Each joint was adjusted to provide ample torque for the unfolding sequence while also staying below the upper limit of the power requirements. The unfolding sequence was repeatedly called, with the adjusted potentiometers the system performed better than initially expected. In testing arm consistently produced enough torque to unfold with TE1 drawing around .9 Amps and TE2 drawing 1 Amps.

Full System Test

Through repeated GSE cycle testing the payload was examined. During testing all inhibit procedures were tested using the inhibit pins. In all combinations of inhibitory the payload control system performed nominally. The GoPro system functioned as expected releasably capturing video in 15 second increments. Upon abrupt power off the system saved video recorded until power loss. The robotic arm was able to move within its usable volume both unlatching and latching successfully.

To adhere to WFF requirements voltage on the pins and plate were repeatedly tested to ensure no voltage was present on the base plate. The center of mass was within regulation, being within one inch from the center of the base plate.

6.0 Mission Results

Following recovery of the payload, both SD Cards were analyzed for mission results and the payload was physically inspected to determine the condition of the mechanical and electrical components.

Post Flight Experiment Condition

Following the flight and recovery, the payload was in relatively good condition. While some evidence of reentry was present most subsystems survived with relatively little damage. During reentry the EMBER system was lost. This was expected outcome considering the mounting points were made of ABS and could have been prevented by milling the housing out of aluminum.

After removing the dry box lid, it was apparent that the seals did not hold. Removing the top plate the team found that all components were submerged in seawater. The silicone RTV utilized to encase both SD cards performed exemplarily well with data being recovered from both cards. Due to the presence of water no parts housed in the dry box will be reused.



Figure 19 – Experiment condition post reentry

Mission Results

After analyzing the SD Cards, students analyzed the video recorded on the GoPro. Early in the footage it becomes apparent that something is partially occluding the lens. Viewing footage from GSE testing at WFF it was determined that no compound was present on the lens until launch. Therefore, it is believed that an unknown compound potentially outgassed and covered the lens during the experiment. This was corroborated by the presence of seawater in the dry box determined upon post launch inspection at Wallops. One of the seals of the dry box did not hold allowing for outgassing to occur while also allowing seawater to wash the unknown compound off the lens leaving students unable to determine the compound. The compounds' effect on video quality can be seen in figure 32. This issue was only compounded by glare from the sun.



Figure 20 – GoPro Footage with Glare.

From the footage itself it is clear that the arm did not unfold as desired. From the audio it is apparent that it attempted to unfold multiple times but was unsuccessful. On startup the encoders did not initialize in the same position as was initially programed in testing. This behavior was not inherently problematic. The arm being fixed at only the first and fifth joint left the other 3 joints to move freely when secured to the payload deck. Due to labor shortage and time constraints a proper calibration system for the arm was not developed. Instead, students determined where the arm would be positioned on spin down and based all joint movements on this assumed position.

The arm's failure to unfold caused another issue with the state machine. The mission control system was programed to signal the EMBER system to deploy objects once the arm had reached its preprogramed catching position. As the arm did not reach this position the computer attempted to refold the arm without launching any deployable. While this system makes sense logically, the EMBER system was never signaled to launch deployable leaving students with no data of its performance in space. In hindsight it would have been best practice to ensure that all subsystems were tested during the mission even if the systems they interact with fail to perform as expected. In the case of EMBER this would involve calling the deployment state in the mission control system after the arm unfolding sequence regardless of the arms success.

7.0 Conclusions

While the results from the launch were not desirable, this year the team collected footage. Also, the lessons learned throughout the development of the system were invaluable.

Choosing to use a GoPro camera as our vision system greatly reduced the team's expected points of failure. Given the correct power requirements this system. This was a great improvement over past teams' issues with vision systems. As the GoPro is a stand-alone unit the team had no need to route data lines from the camera to the dry box. Furthermore, the firmware for the camera was well documented and easy to program compared to other Arduino based options.

The dry box failed to keep the electrical components dry. With the large number of last-minute changes to design details, more holes than desired were drilled into the dry box. These were due to needed mounting points for subsystems using threaded holes or power/data transfer to the main computer. These needs combined added greatly to the number of failure points. While the team is uncertain of the exact failure point of the dry box it is apparent that less points of entry would have increased its chances of holding against a vacuum. Using a single flange and gasket with a high vac connector for power/data would have increased the likelihood of holding against the vacuum of space.

This year's project highlighted the importance of good leadership and proper communication. Most of our design team leads graduated in May leaving important positions in need of filling. While these positions were quickly filled there was a large disconnect between the past team's knowledge and the new team's understanding of the current project. Knowing this now, the team plans on placing more emphasis on the education and training of team members as they begin to fill fundamental team roles.

8.0 Potential Follow-on Work

The team plans to continue iterating this line of research in future years. It was apparent from testing the robotic arm requires a more robust system to effectively meet its mission parameters. One major source of improvement could stem from a proper calibration system for the encoders. Since the magnetic encoders lose their memory on power down, adding a system that could effectively zero the encoder positions in a known arm orientation could greatly improve its accuracy over repeated testing. Current team members have proposed using limit switches to accomplish this calibration system. By moving the arm in a singular direction until it contacts with a set of switches the position of each joint can be determined. The encoders can then be reset with a known constant position for the robot.

Proper inverse kinematics calculations could greatly improve the current system. Having a known relationship between the vector location of the robotic arm in relation to the angles of each articulated joint would allow for proper integration of the vision system. Without such a system it would be impossible to use the vector location returned from the color map. The depth camera could return the location of the ball in relation to the camera, but with no relationship determined between the arm's current vector location and the joint angles, the main computer could not move the arm to the ball's position. The use of a

choreographed position-based system effectively moves the arm, however, the exact nature of the application must be predetermined.

The color-based vision system was only tested using normal ambient lighting in a controlled environment. Building a test environment that mimics that of space could ensure more accurate testing occurred. Adding a lighting system that illuminates the possible deployment field of the ball would ensure the vision system has the best possible chance of success in detection. Furthermore, with recent developments in single shot detectors the team has considered switching to a system that implements object detection as the vision system. Detectors like YOLOv8 have been proven effective on platforms with little computational power while still offering a robust detection system when compared to other models.

9.0 Benefits to the Scientific Community

Although not much data was received during the mission, the team was able to prove a highly capable robotic arm could be developed within small payload volume. This model was strong enough to survive the impulse produced on launch and met strict power requirements.

Far reaching applications can extend to technology that would be beneficial towards manufacturing and servicing spacecraft in orbit and removing space debris.

10.0 Lessons Learned

Throughout the development of the payload and after analysis of the flight data, many lessons have been learned. These lessons will be applied in future years to ensure that the team has the maximum chance of success.

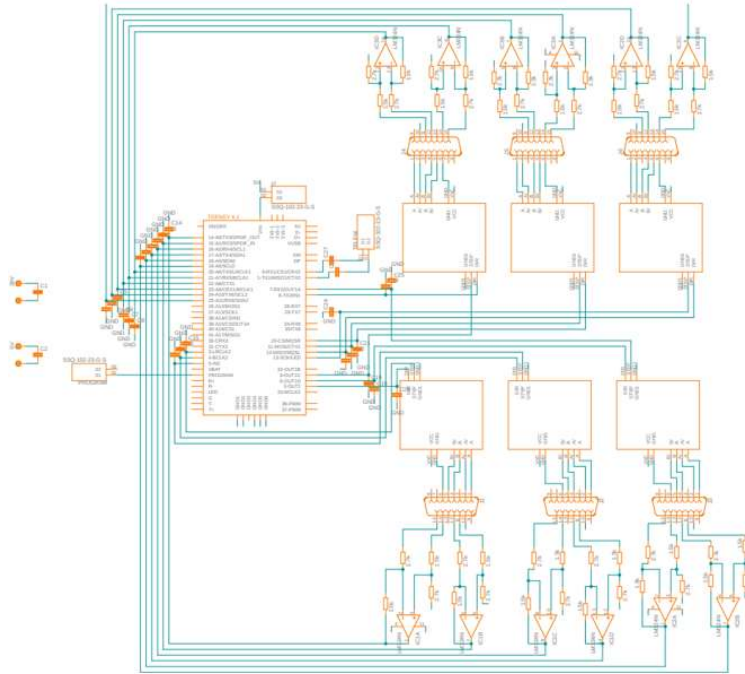
The first and largest lesson learned related to our mechanical design. In developing CAD models a housing was designed for all of the electrical and optical components. It supported the arm and deployment system. This housing was the diameter of the maximum build volume and the height of the max height minus the height of the arm. It was assumed that all electrical components would fit within this space. It was only after multiple members of the team graduated and new roles were assigned that the problem with this assumption was made apparent. The new team realized that components like the GoPro and Nvidia Jetson would not fit within this volume. Due to this the mechanical team had to redesign and move the mounting locations for all subsystems. As many of the subsystems were already finalized the team did not have the option of changing specifications of any singular

subsystem without heavily redesigning. Due to this assumption a significant amount of time spent redesigning. This issue pushed back the team's machining timeline. This led the experiment to be assembled later than desired with time allocated for full mission simulations instead of being used to redesign and machine parts that would fit within the half cam volume. In redesigning the payload more points of failure were added to the dry box likely causing that system's failure. To avoid this issue in the future mechanical designs will be discussed with all members of the team and an advisor. In discussing the design fully, the team is likely to catch dangerous assumptions.

Furthermore, another problem with the final design was caught during machining. The team assumed that fixing the arm at the base and hand would be sufficient to secure it during launch and reentry. While this assumption was true, it left three of the five joints free to move during launch. Had this been caught before the full experiment was assembled, a system could have been added to fix all joints before launch. A burn wire was suggested to secure all joints during launch. While this wouldn't have directly fixed the encoder's memory problem, it would have inhibited movement more than the current system. With a burn wire the encoders would have had a more likely probability of success at zeroing in the same location. Unforeseen problems are bound to occur however the team didn't properly allocate enough time if an issue were to arise. If they had a possible solution could have been implemented before visual verification.

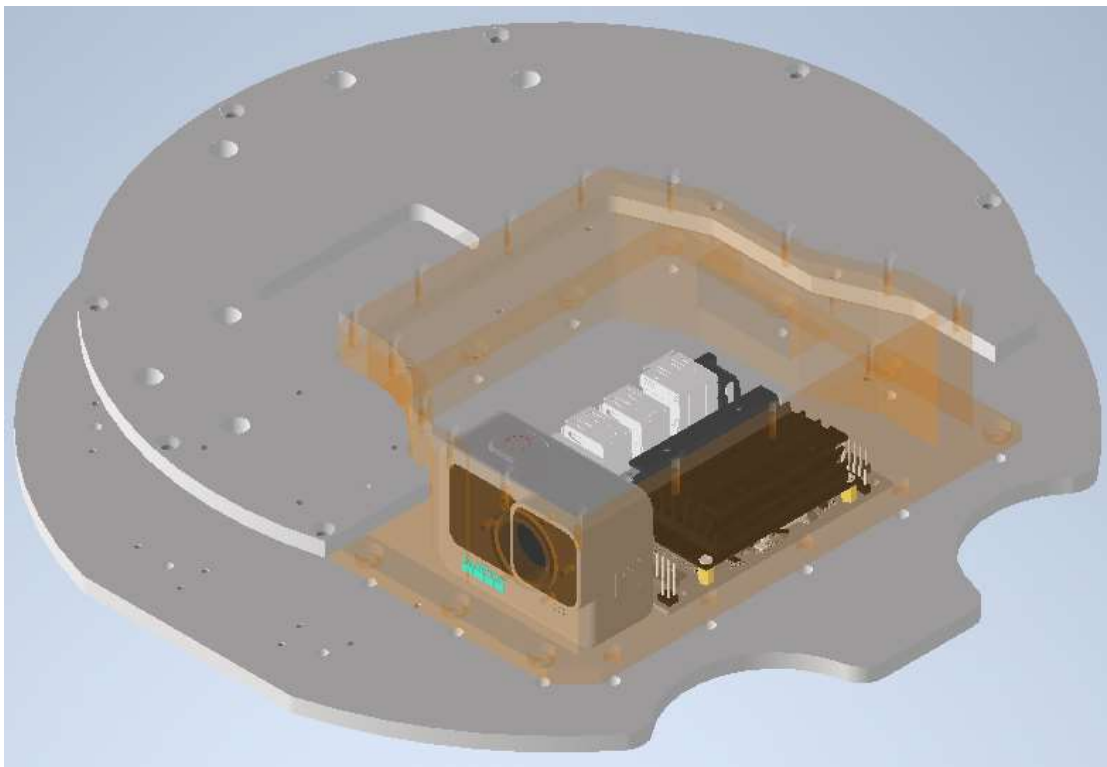
An issue occurred when assembling the final PCB. Unexplainable noise began appearing on the board after the team had applied RTV to the board. The board had interfaced nominally with the Teensy and arm encoders until the back of the board was resealed. After trouble shooting it was determined that something had caused a short between components. After cleaning and reapplying silicone RTV to the board it functioned nominally. The workstation originally used had been used to cut wires and solder components and was not cleaned between processes. The team assumes something like a fine clipping of an encoder wire had attached itself to the board. This could have been avoided by keeping workstations clean

11.0 Appendices



Main PCB Electrical Schematic

Final Board Layout



Transparent Dry Box With Components